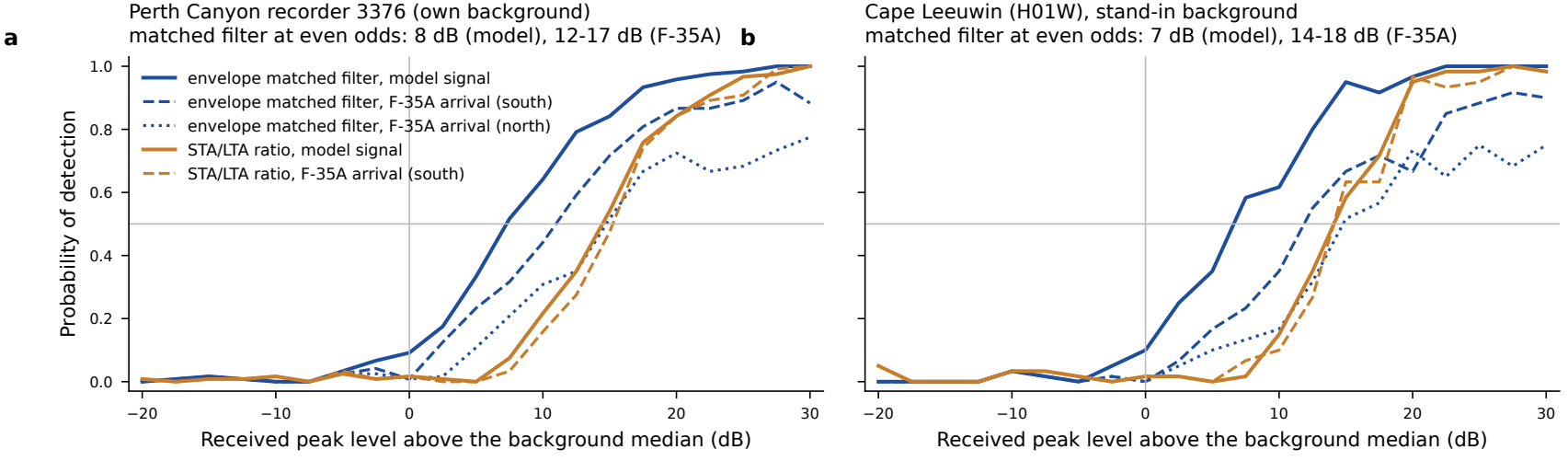
On the real F-35A arrival the matched filter needs 4-10 dB more signal than on the model signal, and gains only 0-3 dB over STA/LTA



(i) We added the sound of the F-35A crash, made again from the published chart, to real background noise at many levels, and counted how often each detector found it. The detector that is tuned to our model signal finds the real crash sound less well than it finds the model signal. Estimates only.

(ii) Pre-registered (prereg 7dc315f) step 3. Arrival = time-frequency random-phase reconstruction of Metz Fig. 3A/B (single elements H11N1, H11S1; DIGITISED), 5-40 Hz; block 7 backgrounds (3376 14-day set; H01W = 3376 shaped to the Blackman 2004 median), detectors, 9-template bank and alpha 0.01 thresholds unchanged; N = 120 (3376) / 60 (H01W) per level; paired model control = block 7 pulse, tau 3.30 s, same files and onsets. Envelope-times-spectrum reconstruction and alpha 0.05/0.001 in the note. Caveats C1 (propagation slope) and C2 (coupling not calibrated for an airliner) apply. All F-35A data DIGITISED from Metz, Obana & Fukao (2023, Pure Appl. Geophys. 180:1343).